

RUPESH GARREPALLI

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PROFESSIONAL SUMMARY

- AI/ML Engineer with 4 years of experience building enterprise machine learning and Generative AI solutions using Python, NLP, and Large Language Models supporting semantic retrieval and inference workflows processing 50K+ daily signals.
- Developed Retrieval-Augmented Generation and embeddings-based semantic search pipelines using LangChain, LlamaIndex, OpenAI API, Azure OpenAI, Pinecone, FAISS, and ChromaDB improving enterprise knowledge discovery accuracy by up to 38%.
- Implemented scalable ML lifecycle pipelines using Scikit-learn, PyTorch, MLflow, Airflow, Docker, and Kubernetes enabling automated feature engineering, experiment tracking, and batch inference across distributed ML environments.
- Deployed cloud AI workloads using AWS SageMaker, AWS Bedrock, Vertex AI, and Azure Machine Learning supporting secure model serving and semantic retrieval across platforms handling 120K+ enterprise records.

TECHNICAL SKILLS

Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, Semantic Search, Embeddings, Feature Engineering, Model Training, Model Evaluation, Model Deployment, Batch Inference, Real-Time Inference, Model Monitoring.

LLM & Generative AI Engineering: LangChain, LlamaIndex, OpenAI API, Azure OpenAI, AWS Bedrock, HuggingFace Transformers, Vector Databases (Pinecone, FAISS, ChromaDB), Context Augmentation, Retrieval Optimization, Hallucination Reduction.

Python ML Ecosystem: Python, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn.

MLOps & ML Platform Engineering: MLflow, Airflow, Model Versioning, Experiment Tracking, CI/CD for ML, Docker, Kubernetes.

Cloud AI Platforms: AWS SageMaker, AWS Bedrock, Google Vertex AI, Azure Machine Learning.

Streaming & Distributed ML Systems: Kafka, Distributed ML Pipelines, Streaming Feature Pipelines, Event-Driven ML Architectures, Microservices-based Model Serving.

Data Engineering for ML: ETL Pipelines for ML Workloads, PostgreSQL, MongoDB, Elasticsearch, Data Modeling for ML Pipelines.

PROFESSIONAL EXPERIENCE

AI/ML Engineer

May 2024 - Present

TCS | USA (Remote)

- Engineered enterprise-scale feature engineering pipelines using Python, Pandas, and PostgreSQL enabling structured transformation of 50K+ daily operational signals supporting predictive intelligence services.
- Developed supervised anomaly detection models using Scikit-learn improving alert prioritization accuracy by 29% across enterprise monitoring analytics workflows supporting platform reliability teams.
- Implemented embeddings-driven semantic retrieval services using HuggingFace Transformers and FAISS enabling contextual enterprise knowledge discovery across structured and unstructured operational datasets.
- Operationalized batch inference pipelines using Airflow and Docker enabling automated nightly model scoring across distributed enterprise analytics environments improving scoring reproducibility.
- Standardized experiment tracking workflows using MLflow improving lifecycle visibility across multi-team enterprise ML deployment environments.
- Optimized feature transformation pipelines using NumPy with Matplotlib and Seaborn validation improving dataset preparation throughput by 34% across enterprise training data readiness workflows.
- Built and owned RAG-based pipelines using FAISS improving enterprise knowledge retrieval accuracy by 15% across distributed semantic search workloads.
- Developed LLM-powered inference workflows using OpenAI and Azure OpenAI reducing enterprise query resolution time by 30% across internal analytics support platforms.

Applied AI Engineer

Jan 2021 - Jan 2023

EdgeVerve | Bangalore, India

- Engineered NLP-driven semantic extraction pipelines using HuggingFace Transformers, TensorFlow tokenization utilities, and Python reducing manual reconciliation validation effort by 41% across enterprise transaction processing workflows.
- Implemented vector similarity indexing architectures using FAISS enabling contextual retrieval across reconciliation document repositories improving entity-matching accuracy across audit intelligence systems.
- Constructed structured feature engineering pipelines using Pandas and PostgreSQL supporting anomaly detection model readiness across transaction monitoring analytics environments.

- Designed event-driven inference enrichment workflows using Kafka enabling streaming contextual intelligence integration across reconciliation decision-support services.
- Operationalized experiment tracking pipelines using MLflow improving lifecycle traceability across supervised learning model iterations supporting compliance analytics governance workflows.
- Containerized transformer inference services using Docker improving runtime portability across distributed reconciliation automation microservices environments.
- Enhanced embedding quality through tokenization tuning strategies improving semantic entity resolution accuracy across enterprise validation workflows.
- Validated model evaluation outputs in collaboration with compliance analytics teams improving explainability confidence across automated reconciliation intelligence pipelines.

PROJECTS

Enterprise Document Intelligence Retrieval Platform

- Engineered embeddings-driven enterprise document indexing pipelines using LangChain, OpenAI API, and Pinecone vector storage enabling semantic retrieval across 120K+ structured and unstructured knowledge records supporting internal analytics discovery workflows.
- Implemented Retrieval-Augmented Generation orchestration using LlamaIndex and Azure OpenAI embeddings improving contextual enterprise document grounding accuracy by 38% across distributed knowledge access queries.
- Operationalized containerized inference services using FastAPI, Docker, and MLflow model lifecycle tracking enabling scalable enterprise semantic retrieval APIs supporting concurrent internal decision-support workloads.

Financial Transaction Semantic Reconciliation Assistant

- Constructed transformer-based semantic entity extraction pipelines using HuggingFace Transformers, TensorFlow tokenization utilities, and FAISS embeddings improving reconciliation entity matching precision by 39% across transaction validation datasets.
- Implemented contextual similarity indexing workflows using vector embeddings and cosine-distance ranking improving reconciliation exception resolution speed by 33% across enterprise audit intelligence pipelines.
- Deployed containerized NLP inference services using FastAPI and Docker enabling scalable semantic reconciliation automation APIs supporting distributed financial validation workflows across internal analytics environments.

EDUCATION

Master of Science in Computer Science

Missouri State University | Missouri, USA

CERTIFICATIONS

- Machine Learning Specialization - **Coursera**
- Deep Learning Specialization - **Coursera**
- Natural Language Processing Specialization - **Coursera**
- Generative AI with Large Language Models - **Coursera**
- Machine Learning Engineering for Production (MLOps) Specialization - **Coursera**
- Generative AI Engineering with LLMs Specialization - **Coursera**